



Rajarata University of Sri Lanka
Faculty of Technology
ICT 2211 - Fundamental of Statistics
Assignment -01

Instructions:

- Please submit your handwritten assignment answers to the ICT Department **on or before 4:00 PM on August 11, 2026.**
- Late submissions **will not be accepted** after the deadline.
- If any form of plagiarism or copying is identified, **a mark of zero** will be given for the assignment

01. Ten patients at a doctor's surgery wait for the following lengths of times (in minutes) to see their doctor:

5, 17, 8, 2, 55, 9, 22, 11, 16, 5

- a) Calculate the mean, median and mode for the above data.
- b) Compute the standard deviation of the data.
- c) Construct the box plot. Are there any outliers in the data?
- d) What measure of central tendency would you use to represent the data? (Mean, median or mode). Give reasons.

02. Mrs. Silva, a mathematics teacher at a local school, recently conducted a mid-term exam for her class of 50 students. She noticed that students had mixed reactions after the exam: some found it easy, while others said it was challenging. To better understand the performance of her students and decide how to adjust her teaching methods, she grouped the students' scores into intervals.

Score Range (Class Interval)	Frequency (Number of Students)
0 - 9	3
10 - 19	5
20 - 29	8
30 - 39	12
40 - 49	10
50 – 59	7
60 – 69	3
70 – 79	2

- e) Find and interpret the average mark
- f) Draw the frequency histogram and relative frequency histogram for the marks of the students
- g) Compare the appearance of the two graphs
- h) Comment on the shape of the distribution
- i) Find the standard deviation of the data above

03. The temperature of two cities A & B in a winter season are given below:

Temperature in City A (in $^{\circ}\text{C}$)	18	25	22	24	20
Temperature in City B (in $^{\circ}\text{C}$)	11	14	16	18	18

- a. Calculate the coefficient of variation for both cities
- b. Find which city has more variability in temperature changes. Give reasons for your answer

04. Mr. Perera is analyzing his monthly household expenses for the past year. He categorized his expenses into intervals and recorded the frequency of months falling into each expenses range

Expense Range (in LKR)	Number of Months (Frequency)
10,000 – 20,000	1
20,000 – 30,000	2
30,000 – 40,000	3
40,000 – 50,000	4
50,000 – 60,000	2

- c. What is the range of the expenses and what does it tell you about the spread of Mr. Perera's monthly spending?
- d. Calculate the average monthly expense of the given data
- e. Find the median and modal class of the monthly expenses
- f. Find the coefficient of skewness and interpret it

05. A random sample of 10 students' heights (in feet) were reported. You have been provided the following information:

$$\Sigma x = 58 \quad \Sigma x^2 = 364$$

Calculate the sample mean, sample variance, and sample standard deviation using the given information